

Manpreet Singh

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SUMMARY

- **Results-driven Analytics Engineer with 5+ years of experience** delivering high-impact data solutions across semiconductor, tech, and research industries. Proven track record of boosting operational efficiency, reducing costs, and enabling data-driven decision-making through advanced analytics, scalable data pipelines, and automation. Skilled in building robust ETL systems, cloud-based architectures (AWS, GCP, Azure), and deploying machine learning models to solve real-world business problems. Adept at collaborating cross-functionally to define KPIs, develop self-serve analytics tools, and lead end-to-end data projects. Technical expertise spans Python, SQL, Spark, Tableau, and modern data stacks, with a strong focus on performance optimization, anomaly detection, and predictive modeling.

EDUCATION

Boston University | Boston | Minor in Data Analytics

Jan 2023

Masters of Science in Computer Information Systems

- Teaching Assistant for Data Science and Data Warehousing & ETL.
- Graduate Research Assistant under Prof. Eugene Pinsky (Project: Automated Portfolio Optimization)

Guru Gobind Singh Indraprastha University | New Delhi

June 2017

Bachelor of Technology in Computer Science Engineering

WORK EXPERIENCE

Analytics Engineer | ASML | Wilton, CT

Feb 2023 - Oct 2024

- Boosted wafer scan output by 19.1% and cut time 15.4% via Python data pipelines.
- Cut costs 30% and cloud spend 16.7% by leading event-driven reporting system.
- Saved \$2M/year by automating diagnostics with 20+ Python scripts.
- Reduced analysis time 30% with interactive dashboards (Altair, Seaborn).
- Improved data quality by implementing robust validation checks, reducing anomalies by 40%.
- Integrated sensor and equipment log into unified data models, enhancing traceability and root-cause analysis.
- Accelerated batch data processing by 2.3x using optimized Pandas and NumPy operations.
- Deployed SQL-based data marts to support self-serve analytics, cutting ad-hoc queries by 50%.
- Collaborated with cross-functional teams to design KPIs for yield loss, driving 12% improvement in defect detection.

Data Science Analyst | Soroco | Bengaluru, India

July 2019 – Aug 2021

- Performed advanced clickstream modeling to identify \$2M in automation opportunities, enhancing prediction accuracy by 45% for Fortune 500 clients.
- Applied process mining to build scalable ML models, enabling 35% cost savings for 10+ Fortune 500 clients.
- Delivered actionable insights through advanced data analysis and visualization.
- Automated data reporting, boosting accuracy and efficiency.
- Translated data findings into clear business recommendations.
- Developed automated data pipelines leveraging MySQL, Python, and Redshift to detect process variations, resulting in \$100K annual savings and 40% reduction in processing time.
- Implemented Isolation Forest-based anomaly detection models to uncover hidden process inefficiencies, facilitating digital transformation initiatives for 5+ clients.
- Defined and monitored key operational KPIs through data-driven analytics, boosting process efficiency by 30% for multiple enterprise clients.

Data Engineer | Shore Infotech India Pvt. Ltd. | Hyderabad, India

Aug 2018 - May 2019

- Built Python-SQL data pipelines, improving extraction efficiency 50% and cutting costs 35%.
- Developed parsing bots for multi-currency financial data with 99% accuracy, automating 35K hours/year.
- Redesigned ETL for 2+ TB financial data, increasing processing speed 60% for real-time analytics.
- Optimized database queries to reduce latency by 40%, enhancing data retrieval speed.
- Implemented data validation frameworks to ensure 99.9% data quality and integrity.
- Collaborated with data scientists to deploy machine learning models into production pipelines.

Data Analyst | Indian Institute of Technology | Kanpur, India

June 2017 – Aug 2018

- Built scalable web scraping tools (Selenium, BeautifulSoup), saving 10,000+ manual hours.
- Delivered 97% accurate forecasts for state electricity grids using ARIMA and Prophet models.
- Automated data pipelines for real-time analytics, cutting latency by 80% across 10+ projects.
- Cleaned and transformed large datasets to improve data quality and drive accurate insights.
- Created interactive dashboards and reports using Tableau and Power BI for data-driven decision making.
- Conducted exploratory data analysis (EDA) to identify trends and support strategic business initiatives.

KEY PROJECTS

Sentiment Analysis of USA Elections by Pyspark

- Ingested 10M+ tweets via Twitter API/Kaggle, processed with Pyspark (text cleaning, deduplication), and stored in Delta Lake for version control.
- Fine-tuned BERT-base-uncased (Hugging Face) for sentiment classification, achieving 85% accuracy using GCP NLP API for comparison.
- Deployed as a serverless pipeline on AWS EMR with Spark Streaming; visualized state-wise trends in Tableau for campaign teams.

Loan Default Prediction using Machine Learning | Boston University

- Performed data mining, clustering, and validation, optimizing detection algorithms and anomaly detection; undertook treatment of missing values.
- Optimized XGBoost, Decision Tree, and SVM models for loan default prediction using cross-validation, achieving 96% accuracy on test data.
- Engineered feature selection pipelines (SHAP, PCA) to enhance model interpretability and accuracy by 10% through dimensionality reduction.

Twilio API Reliability & Data Pipeline Automation

- Built a Datadog dashboard to track SLIs (latency/error rates), reducing downtime by 15% and saving 10K manual hours/year in troubleshooting.
- Programmed ETL scripts for PostgreSQL-to-Snowflake data transfers, boosting efficiency by 25% through automated schema validation.
- Engineered automated Pandas (data-driven) pipelines with Slack alerts, cutting error resolution by 30% while ensuring 99.95% uptime.

Recommender System for Punjabi News Articles

- Developed a content-based recommender system using Bag-of-Words, Word2Vec, and cosine similarity to personalize news recommendations.
- Preprocessed Punjabi text corpus using tokenization, Gurmukhi stop word removal, and stemming, improving recommendation accuracy by 15%.
- Drove 27% higher CTR and 40% curation cost savings by deploying a recommender for Punjabi media, overcoming script challenges.

Automated Asset Allocation for Mutual Funds using Data Science

- Collected and processed data for 10+ mutual funds across 6 categories, engineering 5+ financial features including beta and volatility.
- Built forecasting models (XGBoost, LSTM) achieving >85% directional accuracy in short-term return predictions for top-performing funds.
- Developed asset allocation strategy improving Sharpe Ratio by 22% and reducing portfolio drawdown by 15% compared to static allocation

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn, Altair, Seaborn, Stats models), SQL (MySQL, PostgreSQL, Redshift, T-SQL), R, Java, Bash scripting

Data Analysis & Visualization: Tableau, Power BI, Looker, Qlik Sense, Google Data Studio, Matplotlib, Plotly

Cloud & Big Data Platforms: AWS (S3, Athena, Lambda, EC2, Glue), Snowflake, Google Cloud Platform (Big Query, Dataflow), Microsoft Azure

Data Engineering & ETL: Apache Spark (Pyspark), Delta Lake, Apache Airflow, Kafka, Stream Processing, ETL pipeline design and automation, API Integration (Twilio, Twitter)

Machine Learning & Forecasting: Scikit-learn, TensorFlow, Prophet, ARIMA, Model deployment and monitoring

Automation & Web Scraping: Selenium, BeautifulSoup, RESTful API automation, Workflow orchestration